

GOCHANG, Korea, Republic of

WMO: 471720

Lat: 35.3489N

Lon: 126.5989E

Elev: 184

StdP: 14.60

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.5	19.6	4.6	7.0	26.1	8.2	8.4	27.2	13.4	30.3	12.1	30.4	3.6	0	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.0	91.7	78.3	89.5	77.9	87.3	76.7	80.9	86.4	79.6	85.6	78.4	84.7	6.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.7	155.1	84.0	78.0	146.5	82.9	76.7	140.1	82.2	44.9	86.0	43.4	85.9	42.2	85.2	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.3	11.6	10.0		DB	12.8	95.1	3.7	2.1	10.1	96.7	8.0	97.9	5.9	99.1	3.1	100.6
				WB	11.5	82.4	3.5	2.6	9.0	84.3	6.9	85.8	5.0	87.2	2.4	89.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	31.7	36.2	44.4	54.3	65.0	72.8	79.6	80.3	71.8	60.4	48.8	36.2
	DBStd	17.67	6.72	7.22	6.67	6.43	4.84	3.24	3.88	4.36	4.50	5.56	7.42	7.15
	HDD50	1739	569	391	201	32	0	0	0	0	0	4	112	430
	HDD65	4402	1034	806	638	322	60	0	0	0	4	159	487	892
	CDD50	4263	0	5	29	162	467	685	916	940	653	327	76	3
	CDD65	1450	0	0	0	3	62	235	451	475	206	17	1	0
	CDH74	12053	0	0	3	31	501	1374	4139	4728	1207	70	0	0
	CDH80	3686	0	0	0	2	68	259	1328	1791	234	4	0	0
Wind	WSAvg	3.9	3.6	4.1	4.4	4.4	4.1	3.8	4.3	3.9	3.3	3.2	3.8	4.0
Precipitation	PrecAvg	53.2	1.3	1.7	2.4	3.4	3.7	6.1	11.3	11.8	5.6	2.1	2.2	1.6
	PrecMax	75.3	3.3	4.8	5.0	9.1	7.7	13.9	23.1	21.4	20.3	7.5	6.1	2.9
	PrecMin	31.6	0.4	0.4	0.6	0.7	1.0	1.1	1.7	3.3	0.9	0.2	0.2	0.1
	PrecStd	10.6	0.7	1.1	1.2	2.0	1.6	3.4	4.8	5.1	4.4	1.7	1.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	64.0	70.8	77.3	84.7	87.3	93.3	95.1	88.1	79.6	72.6	60.6
		MCWB	47.4	56.7	54.4	58.1	62.8	69.1	78.5	78.7	75.1	69.3	62.7	53.6
	2%	DB	50.5	57.9	64.9	73.8	81.8	85.0	90.8	92.9	85.2	75.6	67.4	54.5
		MCWB	44.2	51.7	52.4	57.8	62.5	69.2	78.5	78.0	73.5	63.8	59.4	48.8
	5%	DB	45.7	52.4	61.3	70.4	79.2	83.0	88.8	90.9	82.6	73.1	63.9	50.7
		MCWB	40.0	44.6	51.6	55.3	61.8	68.8	77.9	78.2	71.8	62.3	57.2	45.6
	10%	DB	42.1	48.2	57.7	66.8	76.7	81.1	86.8	88.6	80.4	70.4	61.1	47.4
		MCWB	36.9	42.1	49.1	54.2	61.4	68.5	77.3	77.9	70.3	61.5	55.0	42.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	58.7	59.1	63.8	71.4	75.7	83.7	82.7	78.3	73.2	64.6	55.8
		MCDB	52.9	61.9	63.9	72.4	74.7	80.6	88.0	88.8	84.4	76.8	70.4	57.9
	2%	WB	44.9	52.3	56.0	60.7	67.9	74.3	81.3	81.1	76.3	69.0	61.9	49.4
		MCDB	49.9	55.8	61.9	68.5	75.1	79.7	85.7	87.2	81.7	72.7	65.3	53.2
	5%	WB	40.6	46.0	52.9	58.6	65.7	72.9	79.6	80.0	74.5	66.2	58.8	46.2
		MCDB	44.4	51.0	60.0	65.7	74.1	78.5	84.8	86.4	79.6	69.1	63.3	50.1
	10%	WB	37.4	42.3	49.3	56.8	63.9	71.4	78.3	79.0	72.6	63.3	55.6	43.0
		MCDB	41.7	47.9	56.6	63.7	72.4	77.1	84.1	85.7	78.3	68.2	60.1	47.2
Mean Daily Temperature Range	5% DB	MDBR	14.0	15.3	18.8	20.6	20.4	16.0	11.9	13.0	15.5	17.9	15.7	13.4
		MCDBR	19.4	21.9	24.7	26.3	24.6	19.5	15.1	16.6	17.5	20.7	18.3	17.8
		MCWBR	13.6	14.8	14.4	12.9	10.4	7.4	5.6	5.5	7.3	9.7	11.1	13.2
	5% WB	MCDBR	17.1	18.1	20.1	17.7	18.1	12.8	12.1	13.6	13.9	14.0	14.8	16.3
		MCWBR	13.5	14.7	14.0	10.8	9.9	6.8	5.8	5.9	6.9	9.3	10.9	13.4
Clear-Sky Solar Irradiance	taub	0.454	0.492	0.562	0.580	0.597	0.664	0.617	0.564	0.500	0.478	0.465	0.454	
	taud	1.928	1.869	1.731	1.740	1.754	1.690	1.846	1.982	2.088	2.009	1.977	1.937	
	Ebn at Noon	222	230	226	231	229	214	223	233	241	232	217	211	
	Edh at Noon	47	56	69	72	72	77	65	56	48	48	44	44	
All-Sky Solar Radiation	RadAvg	741	1006	1381	1605	1757	1590	1451	1483	1334	1170	805	649	
	RadStd	60	93	125	128	187	141	195	212	128	97	88	51	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	N/A

Nomenclature: See separate page